Local Anchoring or Broader Projection? A Spatial Analysis of Political Manifestos in the Archelec Corpus

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Abstract

This project studies the spatial orientation of political manifestos from the Archelec corpus, a collection of electoral documents from the French Fifth Republic. The main objective is to analyse whether candidates' discourses are mainly rooted in their local electoral territory, or whether they refer more to national and international issues.

To address this question, I built a NLP pipeline based on OCR texts and electoral metadata. First, the documents are loaded from zipped text files and merged with the metadata using their document identifier. Then, named entity recognition is applied with the French medium spaCy model in order to extract location mentions. Since the extracted entities are noisy, especially because of OCR errors and ambiguous named entities, I added a cleaning and validation step. The extracted locations are then geocoded and classified into three categories: local, national/international, and unknown. The local category is defined by comparing the detected locations with the department associated with each document.

The final analysis is based on spatial ratios computed for each manifesto. These ratios make it possible to compare documents, departments and election contexts according to their degree of local anchoring. The results show that spatial references can provide useful information on the way candidates position their discourse. However, the analysis also highlights important limitations, especially the quality of OCR texts, the noise produced by named entity recognition, and the ambiguity of some geographical names.

1 Corpus and Descriptive Statistics

The analysis is based on the Archelec corpus, which contains political manifestos from candidates in elections of the French Fifth Republic. The corpus combines OCR texts with metadata describing the election, the candidate and the geographical context of the document. For this project, I mainly use the text content, the date, the election context and the department associated with each manifesto.

The preprocessing step is quite simple but important for the rest of the pipeline. First, the metadata file is loaded as a table. Then, the OCR texts are retrieved from zipped archives. Each archive can contain several text files, so the script goes through all zip files, extracts the `.txt` files, and stores their content with the name of the source archive and the text file. The document identifier is then extracted from the text file name and used to merge the OCR texts with the metadata. This gives one final table containing both the textual content and the available metadata.

A light text cleaning is applied before the NLP analysis. Line breaks are replaced by spaces, multiple spaces are reduced, and some OCR-related noisy characters are removed. I did not apply a heavy correction of the OCR output, because this could modify the original political text and make the

pipeline less transparent. The goal was rather to keep a reproducible preprocessing step and then handle the remaining noise during the location extraction and classification stages.

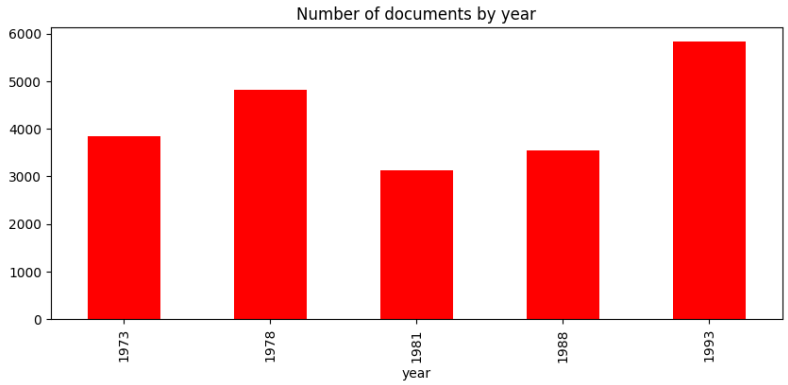

Figure 1: Number of documents by year in the corpus.

Figure 1 shows that the corpus is not evenly distributed over time. The year 1993 is the most represented, followed by 1978, while 1981 contains fewer documents. This matters because any temporal interpretation of the results has to take this imbalance into account. The objective is not to compare perfectly balanced periods, but to describe spatial patterns in the available corpus.

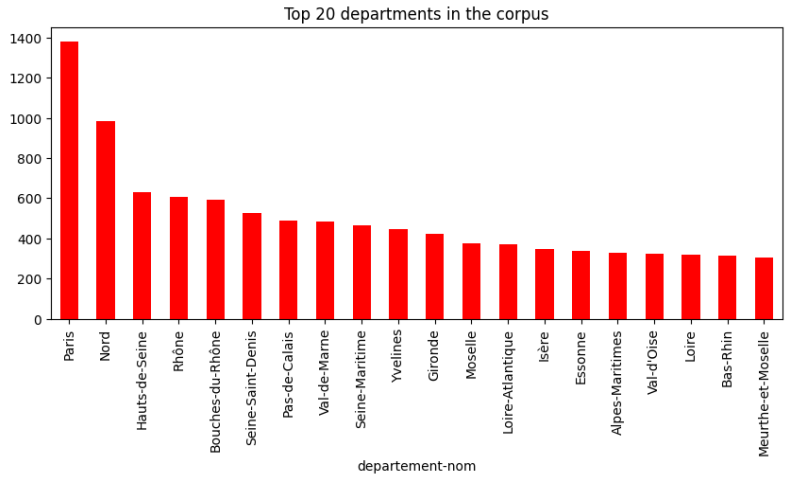

Figure 2: Top 20 departments represented in the corpus.

Figure 2 highlights a strong geographical imbalance. Paris and Nord are clearly the most represented departments, followed by departments such as Hauts-de-Seine, Rhône and Bouches-du-Rhône. This is important for the project because the local dimension is defined at the department level. If some departments are much more present than others, the global results may partly reflect the composition of the corpus.

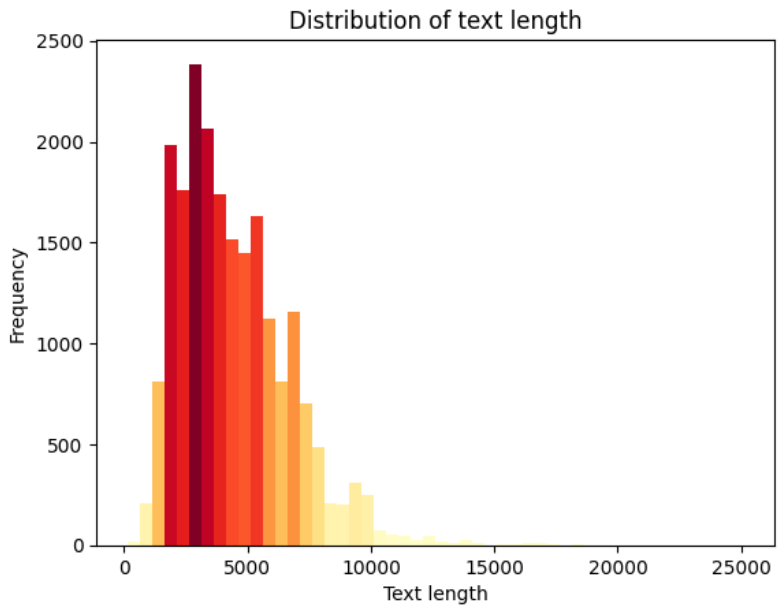

Figure 3: Distribution of document length measured by number of characters.

Figure 3 shows that most texts are relatively short or medium-length, with a concentration around a few thousand characters. There are also some longer documents, which can contain more named entities simply because they contain more text. For this reason, the analysis does not only use raw counts of locations. I compute ratios of local, national/international and unknown mentions for each document, which makes the comparison between documents more meaningful.

Overall, this descriptive step shows that the corpus is usable for spatial analysis, but not perfectly balanced. The main limitations already visible at this stage are the unequal distribution across years and departments, and the variation in text length. These points are kept in mind when interpreting the results.

2 Related Work

This project is connected to three main areas of work: political manifesto analysis, named entity recognition, and geoparsing.

Political manifestos are often used to study how candidates and parties present their priorities during elections. They are not only lists of proposals. They also show how political actors frame problems, justify their position, and speak to voters. In the French case, previous work has used electoral programmes to study the link between campaign promises and policy-making [1]. In my project, I do not analyse ideology or policy promises directly. I focus instead on the spatial dimension of the discourse.

The second part of the project relies on named entity recognition. NER is used to detect entities such as persons, organisations and locations in raw text. I use the French medium spaCy model, `fr_core_news_md`, because it includes a NER component for French texts [2]. This choice is practical and reproducible, but it also has limits. The model is general-purpose and was not trained specifically on OCR political archives. As a result, some extracted entities are noisy or ambiguous.

Finally, the project is related to geoparsing. Geoparsing consists in detecting place names in a text and linking them to real geographical entities. This step is important because a location name alone is not always enough. Some names are ambiguous, and some extracted terms may not be real places. Gritta et al. show that geoparsing requires both toponym detection and careful resolution of place names [3]. This is why my pipeline does not stop at NER. It adds geocoding and a department-based classification in order to separate local references from broader national or international references.

3 Methodology

The core of the project is the pipeline used to transform raw OCR texts into spatial indicators. The idea is simple: extract location mentions from the manifestos, check whether they correspond to real geographical entities, and then decide if they refer to the candidate's local department or to a broader scale.

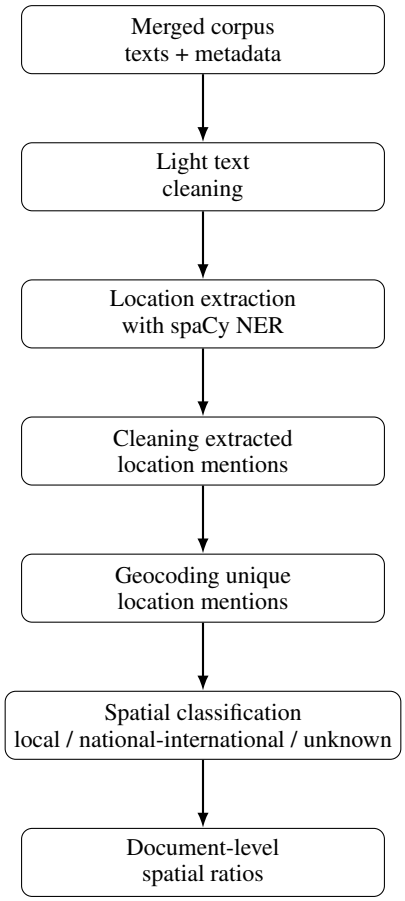

Figure 4: Overview of the spatial analysis pipeline.

Figure 4 summarises the full workflow. The main point is that NER is not used as a final answer. It is only the first step before cleaning, geocoding and classification.

3.1 Text preparation

The merged dataset described in the previous section is used as the starting point. At this stage, what matters is to have the OCR text and the department metadata in the same table. The department is essential later because it is used as the reference territory for the local classification.

The cleaning is intentionally limited. I remove line breaks, repeated spaces and some noisy OCR characters, but I do not try to rewrite or correct the full text. This choice keeps the pipeline transparent. It also avoids modifying old political documents too much before applying NLP methods.

3.2 Location extraction

I use named entity recognition to extract location mentions from the texts. The final model used in the project is `fr_core_news_md`, the medium French model from spaCy. I also tried the small French model, but the results were noisier, with more terms detected as locations even when they were not useful for the analysis.

For each document, the model returns entities labelled as locations or geopolitical entities. These mentions are stored in a column called `locations`. This gives a first view of the spatial references in each manifesto.

The output is still imperfect. Some extracted mentions are clearly valid places, for example *Paris*, *Metz* or *Pas-de-Calais*. Others are more problematic, especially because the corpus comes from OCR texts. This is why the extracted list is not analysed directly.

3.3 Cleaning and validating location mentions

After the NER step, I clean the extracted mentions with a small set of automatic rules. First, I remove line breaks inside location names and strip unnecessary spaces. I also remove French articles or prepositions at the beginning of a mention, such as *le*, *la*, *les*, *de*, *du* or *des*. For example, a mention like *la France* is normalized before the next steps.

I also remove very short mentions, purely numeric values, and a few generic words that were often detected as locations but are not useful for the analysis, such as *gouvernement*, *mesures* or *aide*. This cleaning step remains simple on purpose. It does not try to solve all errors produced by the NER model, but it removes the most obvious noise before geocoding.

The cleaned mentions are stored in the column `locations_clean`.

3.4 Geocoding

The next step is to connect extracted mentions to real geographical entities. To make the process more efficient, I first create a reference table containing only unique location mentions. Each unique mention is geocoded once, instead of being geocoded again for every document.

I use `geopy` with Nominatim. The geocoding returns information such as the country, city, department, region and coordinates. Since the corpus is French, the search is first biased toward France. If no French result is found, the query is then searched more broadly. This helps avoid some obvious errors, for example a French place name being matched with a place abroad.

The output of this step is the table `locations_reference`. It is useful for the automatic classification, but also for checking errors manually during the analysis.

3.5 Spatial classification

The main classification has three categories:

- **local**: the mention refers to the same department as the document;
- **national/international**: the mention refers to a place outside the document's department, or to a broader scale;
- **unknown**: the mention is too noisy or uncertain to be used safely.

The local label is based on the metadata columns `departement-nom` and `departement-insee`. If a geocoded place belongs to the same department as the manifesto, it is classified as local. If it is a valid place but outside this department, it is classified as national/international.

I also keep some territorial markers that are not always places in a strict geographical sense, but still give useful information. For example, words like *circonscription*, *commune* or *canton* are treated as local signals. Terms like *Europe*, *États-Unis*, *national* or *international* are treated as broader-scale signals. This makes the classification closer to the actual research question.

The `unknown` class is important. Without it, the pipeline would force noisy NER outputs into meaningful categories, which would bias the results. Keeping this category makes the analysis more cautious.

3.6 Spatial ratios

The last step is to aggregate the classified mentions at the document level. For each manifesto, I compute three indicators:

- `local_ratio`: share of local mentions;
- `national_international_ratio`: share of broader-scale mentions;
- `unknown_ratio`: share of mentions that could not be classified reliably.

I also compute a dominant spatial scale for each document. This corresponds to the most frequent valid category. These ratios make documents easier to compare, even when they have different lengths or different numbers of extracted locations.

In the end, the pipeline does not only produce extracted place names. It produces document-level indicators that can be compared by year, department or election context.

4 Experiments and Results

This section presents the results obtained with the full pipeline described above. Because the NER + geocoding pipeline is quite heavy and takes several hours to run, the experiments reported here are based on a sample of 500 texts. The goal is therefore not to claim a final result for the whole Archelec corpus, but rather to show the first patterns that appear when the method is applied on a realistic subset.

4.1 Inspecting intermediate outputs

Before looking at aggregated results, it is useful to look at the intermediate tables produced by the pipeline. This makes the process easier to follow and also helps to understand where some errors can come from.

	location	country	country_code	county	state	display_name	geo_class	geo_type
0	Seine-Maritime	France	fr	Seine-Maritime	Normandie	Seine-Maritime, Normandie, France métropolitaine, France	boundary	administrative
3	Agen	France	fr	Lot-et-Garonne	Nouvelle-Aquitaine	Agen, Lot-et-Garonne, Nouvelle-Aquitaine, France métropolitaine, 47000, France	boundary	administrative
4	Libos	France	fr	Lot-et-Garonne	Nouvelle-Aquitaine	Libos, Monsempron-Libos, Villeneuve-sur-Lot, Lot-et-Garonne, Nouvelle-Aquitaine, France métropolitaine, 47500, France	place	hamlet
7	Avesnois	France	fr	Nord	Hauts-de-France	Neuville-en-Avesnois, Avesnes-sur-Helpe, Nord, Hauts-de-France, France métropolitaine, 59218, France	boundary	administrative
9	Lille	France	fr	Nord	Hauts-de-France	Lille, Nord, Hauts-de-France, France métropolitaine, France	boundary	administrative
10	Creuse	France	fr	Creuse	Nouvelle-Aquitaine	Creuse, Nouvelle-Aquitaine, France métropolitaine, France	boundary	administrative
11	Creusois	France	fr	Creuse	Nouvelle-Aquitaine	Rue du 1er Maquis Creusois, Le Moulin, Maisonnisses, Guéret, Creuse, Nouvelle-Aquitaine, France métropolitaine, 23150, France	highway	tertiary
12	Corrèze	France	fr	Corrèze	Nouvelle-Aquitaine	Corrèze, Nouvelle-Aquitaine, France métropolitaine, France	boundary	administrative
13	Millevaches	France	fr	Corrèze	Nouvelle-Aquitaine	Millevaches, Ussel, Corrèze, Nouvelle-Aquitaine, France métropolitaine, 19290, France	boundary	administrative

Figure 5: Sample of the `locations_reference` table after geocoding.

Figure 5 shows a sample of the reference table built after geocoding the unique extracted locations. This table is central in the pipeline because it links each mention to structured geographical information such as the country, county, state, display name, and geocoding type. It also makes some errors visible. For example, clear place names such as *Seine-Maritime*, *Agen* or *Lille* are resolved correctly, while more ambiguous terms can lead to noisier matches. This confirms that the geocoding step is useful, but also that it cannot be trusted blindly.

	departement-nom	locations_clean	classified_locations	local_ratio	national_international_ratio	unknown_ratio	dominant_scale
13775	Seine-Maritime	[Seine-Maritime, France, France, République Française]	[{'location': 'Seine-Maritime', 'scale': 'local'}, {'location': 'France', 'scale': 'national_international'}, {'location': 'France', 'scale': 'national_international'}, {'location': 'République Française', 'scale': 'national_international'}]	0.250000	0.750000	0.000000	national_international

Figure 6: Sample of the final document-level output with classified locations and spatial ratios.

Figure 6 shows the final output at the document level. For each text, the cleaned location mentions are classified and then aggregated into three ratios. In the example shown here, the document is linked to the department Seine-Maritime. The mention *Seine-Maritime* is classified as local, while *France* and *République Française* are classified as broader-scale references. The final dominant scale of the document is therefore `national_international`. This example makes the logic of the pipeline quite concrete.

4.2 Global spatial patterns

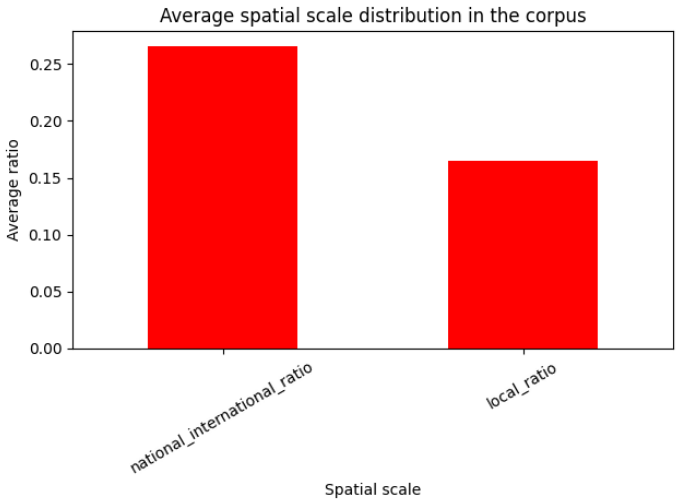

Figure 7: Average spatial scale distribution in the sample.

Figure 7 gives a first overall result. On average, the national/international ratio is higher than the local ratio. This result is partly influenced by the frequent presence of the word *France*, which is classified as a broader-scale reference. I chose not to remove it, because it is still meaningful for the research question. When a candidate refers to France, the discourse is no longer limited to the local department. It reflects a national framing of the political message.

So, even if this category is partly driven by frequent national references, the result remains useful. It suggests that the manifestos in this sample do not stay mainly at the department level. Local references are present, but they are not the dominant tendency on average.

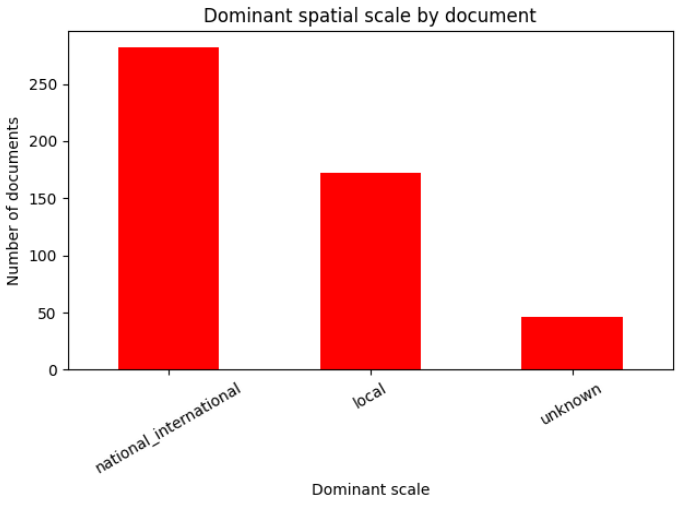

Figure 8: Dominant spatial scale by document.

Figure 8 goes in the same direction. Most documents are dominated by the `national_international` category. A smaller but still important group is dominated by the local category, and only a limited number of documents are dominated by `unknown`. I find this result rather coherent. It suggests that the method is able to capture meaningful spatial signals, while still keeping uncertain cases separate.

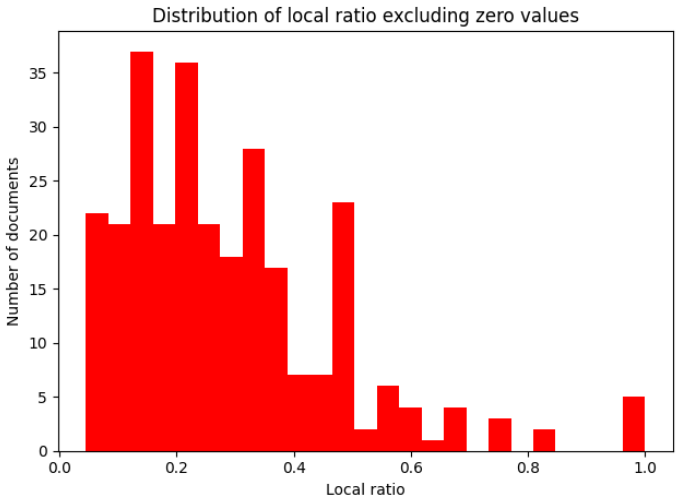

Figure 9: Distribution of local ratio excluding zero values.

Figure 9 focuses on documents that contain at least some local signal. Most of them have a local ratio between roughly 0.1 and 0.4. So even when a manifesto contains local references, these references usually do not occupy the whole discourse. Only a small number of texts appear to be very strongly anchored in the local scale.

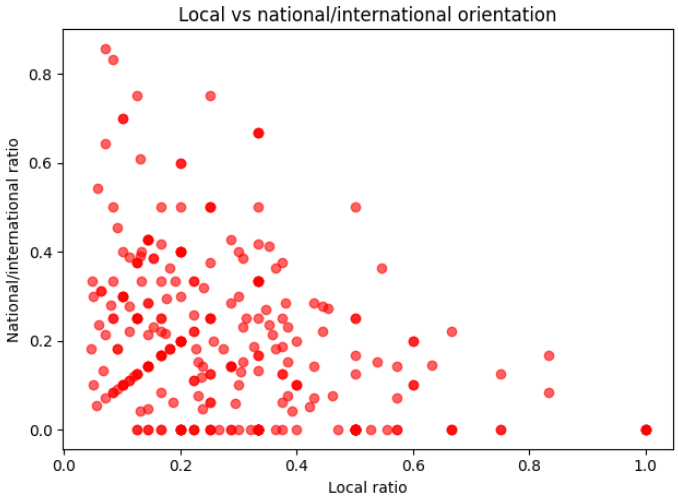

Figure 10: Local vs national/international orientation.

Figure 10 is useful because it shows that local and broader-scale references are not simply opposites. Many documents combine both. A text can contain a clear local anchoring and still discuss France, Europe or international issues. This is actually one of the interesting results of the project. The spatial structure of a manifesto is not binary. It is often mixed.

4.3 Variation over time and electoral context

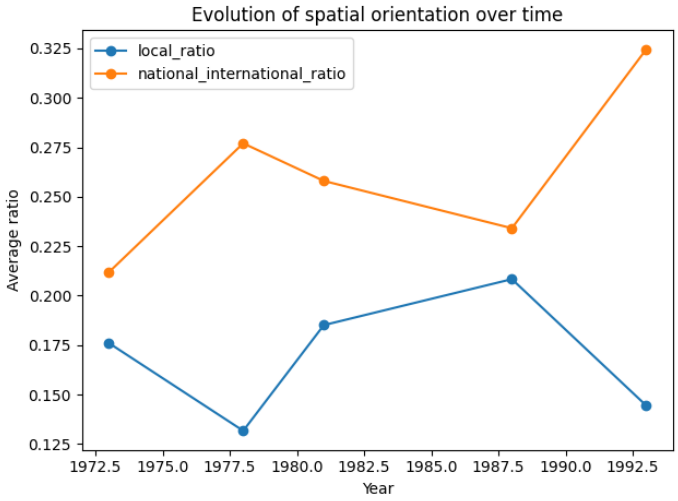

Figure 11: Evolution of spatial orientation over time.

Figure 11 shows the average local and national/international ratios by year. In this sample, the broader-scale ratio stays above the local ratio for all years. There are some variations, with a stronger national/international orientation in 1978 and especially 1993, while the local ratio reaches its highest value around 1988. I would still stay cautious here, because the corpus is not perfectly balanced across years. This figure is interesting as a first descriptive trend, but not as a strong historical claim.

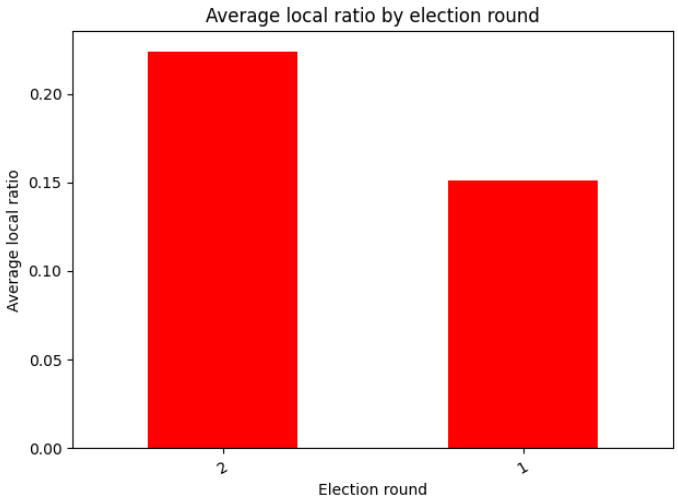

Figure 12: Average local ratio by election round.

Figure 12 compares the average local ratio between election rounds. In this sample, second-round manifestos appear slightly more local than first-round ones. One possible interpretation is that, in the second round, candidates may insist more on territorial anchoring and local legitimacy. Still, I would treat this as an exploratory result rather than a definitive conclusion, especially given the sample size.

4.4 Main takeaway

Taken together, these results suggest that the manifestos in the sample are not purely local texts. The local scale is present, sometimes quite clearly, but broader national and international references are

very common and often dominant. At the same time, the scatter plot and the example tables show that the discourse is often mixed rather than purely one-sided.

For me, the most useful point is that the pipeline produces interpretable document-level indicators from noisy OCR texts. Even on a limited sample, it already shows meaningful differences in spatial orientation. The next step would be to scale the analysis to a larger share of the corpus and to refine the handling of ambiguous location mentions.

5 Discussion

The results are still exploratory, since the pipeline was run on a sample of 500 texts. Running it on the full corpus would take much longer, mainly because of the geocoding step. Still, the sample is useful to check if the method works and if the indicators are interpretable.

The main result is that the discourses are not purely local. The local dimension appears in many documents, but the national/international category is often higher. This is partly explained by the frequent presence of words such as *France* or *République Française*. I kept these terms because they are meaningful for the question: referring to France already means that the discourse goes beyond the department.

Another point is that local and broader references often appear together. Some texts mention local places, while also referring to France, Europe or international issues. So the spatial orientation is not always binary. A manifesto can be locally anchored and still project itself toward a wider political scale.

The main limitation is the quality of the extraction. The OCR is noisy, and the spaCy model sometimes detects words that are not real locations. Geocoding also creates errors when a place name is ambiguous. This is why I used the `unknown` category, instead of forcing every mention into a class. It makes the analysis less complete, but more cautious.

6 Conclusion

This project studied the spatial orientation of political manifestos in the Archelec corpus. The goal was to see whether the discourse is mainly local, or whether it refers more to national and international spaces.

To do this, I built a pipeline using spaCy NER, cleaning rules, geocoding, and a department-based classification. The final output is not only a list of places, but a set of ratios for each document: `local_ratio`, `national_international_ratio`, and `unknown_ratio`.

On the sample of 500 texts, the results suggest that broader references are often dominant, even when local mentions are present. This shows that candidates do not only speak about their local territory. They often connect it to France, Europe, or wider political issues.

The method has limits, especially because of OCR noise, NER errors, and ambiguous place names. A next step would be to run the pipeline on the full corpus and improve the location validation step, possibly with a small manually annotated dataset.

References

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